2. Gas (6 points) — *Jaan Kalda.*

A box of volume V contains ν moles of a monoatomic gas of molar mass μ at a negligibly low temperature. The box stops instantaneously after having moved with a constant speed v (much greater than the thermal speed).

i) *(2 points)* Find the temperature inside the box after thermalization.

ii) *(2 points)* Find pressure to the front wall (the wall that was at the front while the box was moving) immediately after stopping.

iii) *(2 points)* Now, a spherical ball of gas (helium, $\mu = 4\,\mathrm{g/mol}$ of radius $r = 1\,\mathrm{cm}$, at temperature $T = 300\,\mathrm{K}$ is surrounded by vacuum. The mean free path of the molecules is much larger than r. At a certain moment, the walls of the ball break and after $\tau = 5\,\mathrm{ms}$, a part of the gas is trapped by erecting instantaneously walls forming a cubical vessel of volume $V = 1\,\mathrm{m}^3$. Find the temperature of the gas inside the cube T' after thermalization; neglect the thermal capacitance of the walls of the cubical vessel. $R = 8.31\,\mathrm{J/kg\,K}$.